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Project Report

Over the past few decades, artificial intelligence has greatly expanded its applications across multiple industries and fields. Rapid advancements and widespread support for deep learning and machine learning have consistently led to breakthroughs in multiple domains. The possibilities are almost endless. Artificial intelligence is arguably the one field that can help others surpass their limits. Two major areas where machine learning continues to help advance are environmental engineering and business. This project examines how machine learning models can help in these areas by examining datasets on energy efficiency and wine quality.

The rise of climate change has sparked a shift in how people around the world view unchecked energy use. A major problem tackled by the field of environmental science is building physics and how the design of buildings affects energy efficiency. Building necessities such as heating, ventilation, and air conditioning, contribute to roughly 33% of CO₂ emissions and 40% of global energy usage. Further studies and simulations show that a building's operational stage makes up about 85.4% of the total CO₂ emissions over a 50 year time span, starting from construction to demolition. (Peng)

The first dataset, energy efficiency, contains data from simulations conducted on 768 building shapes and designs created with the software Ecotect. The first dataset of this project, energy efficiency, aims to apply machine learning models to the specifications of a building's structure and its thermal performance. To attempt to predict the heating load and cooling load necessary to maintain the interior temperature.

The energy efficiency dataset outputs these two separate target variables, heating load and cooling load, against eight features relating to a building's specifications. The dataset's CSV file lacks clear labeling for all of its inputs and outputs. Six features are continuous inputs and two are integer inputs each numbered from X1 to X8. In order, these features are: relative compactness, surface area, wall area, roof area, overall height, orientation, glazing area, and glazing area distribution. The integer variables orientation and glazing area distribution represent categorical features. Orientation refers to the building's four cardinal points of a building's orientation and the glazing area distribution feature describes the six types of glass distribution for windows. Each glazing area describes the percentage of the glazing areas for each side of a building. The two output variables, Y1 and Y2, stand in for heating load and cooling load, respectively.

The introductory paper for the Energy Efficiency dataset, Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools, used two regression models: Iteratively Reweighted Least Squares (IRLS) and Random Forests. IRLS is a variation of linear regression that accounts for outliers in its calculations, while Random Forest is a far more complex model capable of capturing non-linear relationships in the data. They treated linear regression as a baseline and demonstrated that Random Forest offered far more accurate results compared to simulation outputs. They recommended using Random Forest over simpler models like linear regression for better generalization to similar datasets. The study used 10-fold cross-validation repeated 100 times with different random seeds to ensure reliable evaluation. Instead of the R^2 metric, they reported mean absolute error (MAE), mean squared error (MSE), and mean relative error (MRE). Using mutual information and Spearman rank correlation, they identified that the first five input features had the strongest influence on the output variables. Their models predicted the heating load relatively accurately, but struggled more with the cooling load. Overall, the authors advocated for the use of machine learning in energy efficiency analysis, while cautioning against relying on simple models like linear regression, which may fail to capture the complexity of the data.

For businesses, determining what will be a commercially successful product often seems nebulous. The quality of wine, for instance, often relies on human senses which can be subjective. This project's second focus is to apply machine learning to a variety of red wine offered in Vinho Verde. This data only contains the physicochemical properties and their quality determined on a scale of 0 to 10. By applying machine learning to this second dataset, the target is to predict a wine's quality level based solely on these objective, physicochemical properties.

In the wine dataset, there are a total of 11 input features and one output feature. The 11 input features represent various physicochemical attributes while the final feature is the wine quality assessment rating. Unlike the numbered features in the energy efficiency dataset, all red wine features are given their descriptive names as labels in the CSV file. The continuous physicochemical attributes range from fixed acidity to alcohol content.

The datasets require a bit of preprocessing to ensure data quality. Since we're using regression models, standardization is necessary. Also, the dataset is multivariate meaning that even though all numbers are numeric, there is a combination of continuous and integer features. The dataset is considered complete so it contains no missing values and so no data cleaning is required.. The features require scaling as well as encoding before training the models. The two integer features, orientation and glazing area distribution, in the Energy Efficiency dataset have to be encoded with one-hot encoding since there are more than two classes with no specific correlation with each other.

Cross-validation helps when comparing the performance of different models. When comparing the scores of the models, the goal is to find the model with the best performance, most consistent performance, and reasonable assumptions. Where lower bias and variance are desired. This project does not follow the amount of folds set that the introductory paper suggested. This project uses 5 folds across both datasets.

This project applies three different machine learning regression models, linear regression, k-nearest neighbors regression, and elastic net regression, to both datasets to predict the target variables as accurately as possible. Linear regression is the least complex model and works best when there is a clear linearity in the relationship between input and output variables. Because of this, linear regression is expected to perform well on the Energy Efficiency dataset since the input features will most likely have a linear relationship with the heating and cooling loads.

K-nearest neighbors regression examines the distance between the instances and predicts what the output is based on the average output value from the neighbors with the smallest distances. This model may work well on the Energy Efficiency dataset since it contains less than a thousand instances. This works in K-nearest neighbors favor since the model tends to do well on smaller datasets due to scalability issues.

Elastic net regression is a combination of LASSO and ridge regression. This introduces regularization into its calculations meaning that depending on its hyperparameters, the elastic net model shrinks or flattens weights to zero. The Energy Efficiency dataset is expected to benefit from the model. Elastic net is more stable when there's many features that correlate with each other like the surface area, wall area, roof area, and overall height.

Hyperparameters are user-defined parameters set before a model is trained. Linear regression is a linear model without any hyperparameters. K-nearest neighbors regressor is non-linear with three hyperparameters: `n_neighbors`, `metric`, and the power parameter for minkowski distance. `n_neighbors` determines the number of nearby instances that the model accounts for when calculating the output value. Elastic net has two hyperparameters, `alpha` and `l1_ratio`. `Alpha` controls the regularization strength which, in turn, defines the magnitude that weights to shift closer to zero, reducing the prediction variance. The `L1 ratio` is any value between 0 and 1. When close to zero, elastic net turns into ridge regression which shrinks weights towards zero without ever completely becoming zero. When `L1` is closer to 1, the weights are more likely to become set at zero.

`GridSearchCV` was used during model tuning to determine the best hyperparameters for each model. This method requires the hyperparameter search space to be defined beforehand.

However, examining the validation curve revealed that the training scores were much higher than the validation scores, with a significant gap between the two. This suggests that the KNN model was overfitting to the training data. The overfitting likely

stems from the relatively small number of instances in the Energy Efficiency dataset, which limits KNN's ability to generalize effectively.

Further testing with higher `n_neighbors` values showed that setting `n_neighbors` to around 21 reduced overfitting, with training and validation scores beginning to overlap more closely. Importantly, both scores remained relatively high, around 0.9, indicating that the model was still performing well without sacrificing accuracy. The validation scores across different hyperparameter values remained relatively close to one another, suggesting that the model's performance was fairly stable across different `n_neighbors` settings.

These are the results of this project's tests on the Energy Efficiency Dataset:

Baseline (no tuning):

Least Squared Linear Regression metrics:

R²: 0.9177429554254065

RMSE: 2.7862549812251043

MAE: 1.9657228970507492

MSE: 7.763216820401707

KNN Regression metrics:

R²: 0.9375437493739922

RMSE: 2.4232937068400084

MAE: 1.7670259740259748

MSE: 5.8723523896103895

Elastic Net metrics:

R²: 0.8319873872107597

RMSE: 3.9961011805555415

MAE: 3.070589035827072

MSE: 15.968824645237394

Tuning grid for GridSearchCV:

k-nearest neighbors: {'n_neighbors': [3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25]}

elastic net regression: {'alpha': [0.1, 0.3, 0.5, 0.7, 0.9, 1, 2], 'l1_ratio': [0.1, 0.3, 0.5, 0.7, 0.9]}

Best hyperparameters for KNN:

Mean testing scores: According to the GridSearchCV results, the best hyperparameter value is: KNeighborsRegressor(n_neighbors=9)

Best hyperparameters for Elastic Net:

According to the GridSearchCV results, the best hyperparameter value is: ElasticNet(alpha=0.1, l1_ratio=0.9)

Validation Scores:

Mean: 0.897

SD: 0.015

Mean: 0.931

SD: 0.01

Mean: 0.891

SD: 0.012

This is the results after tuning:

Least Squared Linear Regression metrics:

R²: 0.9177429554254065

RMSE: 2.7862549812251043

MAE: 1.9657228970507492

MSE: 7.763216820401707

KNN Regression metrics:

R²: 0.9648176461647943

RMSE: 1.8254622345392515

MAE: 1.3222438672438677

MSE: 3.3323123697290375

Elastic Net metrics:

R²: 0.90795599670103

RMSE: 2.9462594306141416

MAE: 2.1183787798996496

MSE: 8.680444632482766

After fitting, validating, and testing the final models, the metrics of the linear regression, k-nearest neighbors regression, and elastic net regression models in this project were generally consistent with the results reported in the introductory paper of the Energy Efficiency dataset. Unlike the paper, which reported separate results for the

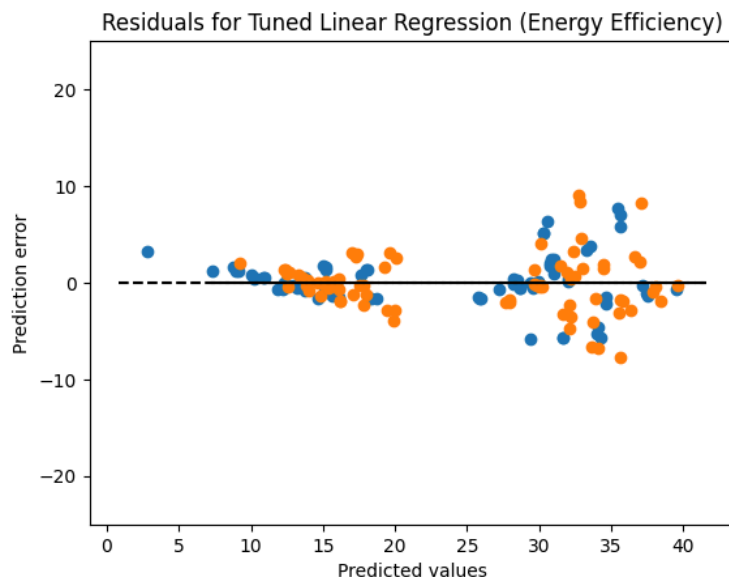
two output variables (heating load and cooling load), this project aggregated the results across both outputs for easier comparison.

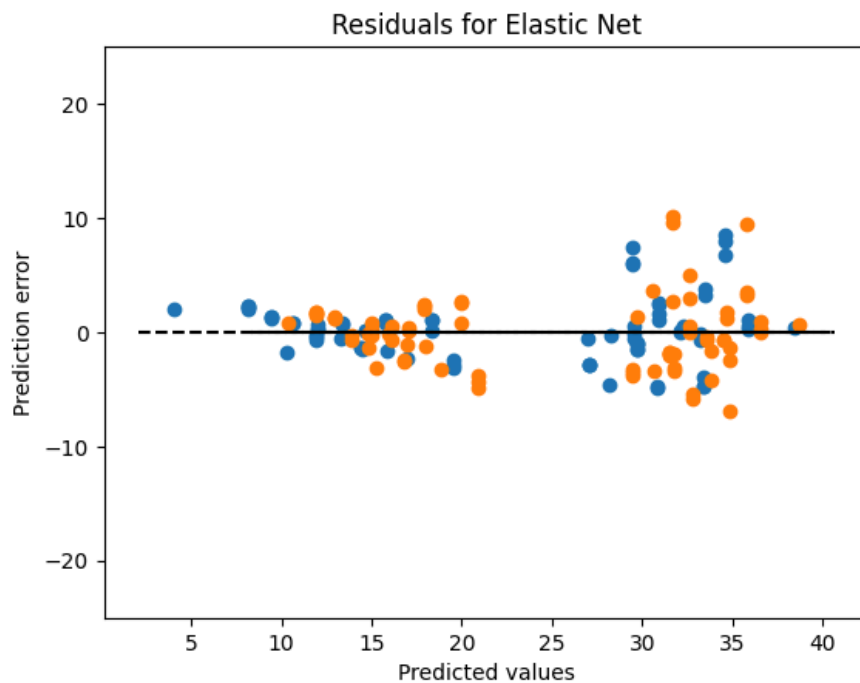
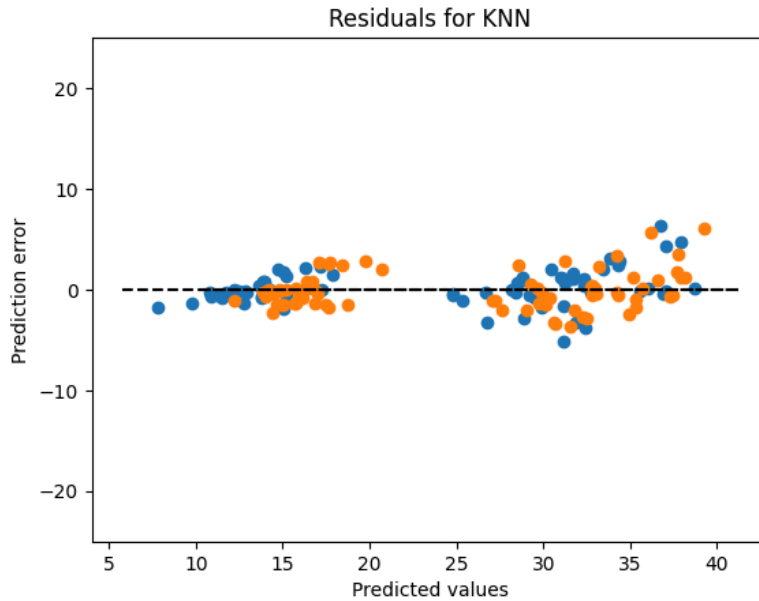
For the mean absolute error (MAE), the introductory paper reported values of 2.14 for heating load and 2.21 for cooling load using linear regression, resulting in an aggregate MAE of 2.175. This project's linear regression model achieved an MAE of 1.966, which is slightly better but still comparable to the introductory paper's results.

For random forests, the introductory paper reported MAE values of 0.51 for heating load and 1.42 for cooling load, with an aggregate MAE of 0.965. In this project, the k-nearest neighbors model, which served as the most comparable model tested, achieved an aggregate MAE of approximately 1.322, which, while higher, is still relatively close.

For the mean squared error (MSE) metric, this project obtained values of 7.763 for linear regression, 3.332 for k-nearest neighbors, and 8.680 for elastic net. In comparison, the introductory paper reported an aggregate MSE of 10.665 for linear regression and 3.81 for random forests. While their random forest results significantly outperformed their linear regression results, this project's k-nearest neighbors model achieved an MSE of 3.332, outperforming the random forest model reported in the paper.

These are the prediction error scatter plots for the models tested on the Energy Efficiency dataset:





The prediction error versus predicted values charts for this project showed similar patterns across all three models. Most errors were centered around a mean of zero, indicating generally unbiased predictions. However, as the predicted values increased, the magnitude of the prediction errors also tended to increase. K-nearest neighbors regression performed the best overall, maintaining consistently low prediction errors across the range. In contrast, both linear regression and elastic net models exhibited a

wider spread of errors at higher predicted values. This suggests that these models struggle with larger predictions on this dataset compared to k-nearest neighbors.

Here are the results for the Red Wine dataset:

Results before tuning:

Least Squared Linear Regression metrics:

R²: 0.2737888679028764
RMSE: 0.7227849840390767
MAE: 0.5648966968438002
MSE: 0.5224181331523683

KNN Regression metrics:

R²: 0.26012163336229366
RMSE: 0.7295546586788408
MAE: 0.57125
MSE: 0.53225

Elastic Net metrics:

R²: -0.0025078585649389584
RMSE: 0.8492226390971644
MAE: 0.7084343236903832
MSE: 0.7211790907551529

GridSearchCV tuning grid:

k-nearest neighbors: = {'n_neighbors':[3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25]}
elastic net regression: 'alpha': [0.1, 0.3, 0.5, 0.7, 0.8, 0.9, 1, 10], 'l1_ratio': [0.1, 0.3, 0.5, 0.7, 0.8, 0.9]

According to the GridSearchCV results, the best hyperparameter value is:

KNeighborsRegressor(n_neighbors=25)

According to the GridSearchCV results, the best hyperparameter value is:

ElasticNet(alpha=0.1, l1_ratio=0.1)

Validation Scores:

Mean: 0.359
SD: 0.064
Mean: 0.337
SD: 0.034
Mean: 0.357
SD: 0.052

These are the results after tuning:

Least Squared Linear Regression metrics:

R²: 0.2737888679028764
RMSE: 0.7227849840390767
MAE: 0.5648966968438002
MSE: 0.5224181331523683

KNN Regression metrics:

R²: 0.26843440486533454
RMSE: 0.7254446912067108
MAE: 0.56425
MSE: 0.52627

Elastic Net metrics:

R²: 0.27237976087975835
RMSE: 0.7234858737509142
MAE: 0.5679249230494767
MSE: 0.5234318095171238

Overall, all models performed significantly worse on the Red Wine Quality dataset compared to the Energy Efficiency dataset. Before hyperparameter tuning, the elastic net model performed the worst, with its lowest RMSE at 0.849. Although the raw error metrics (such as RMSE and MAE) were numerically smaller than those in the Energy Efficiency dataset, the scale of the target variables must be considered. Heating load and cooling load values often reach up to 50 kWh, while the maximum quality score in the Red Wine Quality dataset is only 10. Thus, relative to the range of the target values, the models show much poorer performance on the Red Wine dataset across all metrics.

The results across the tuned models were extremely close. The more complex models, k-nearest neighbors and elastic net, performed similarly to linear regression. This suggests that there is little to no strong correlation between the given physicochemical attributes and the critical reception of the wine. The validation curves also present clear evidence of this: both the k-nearest neighbors and elastic net validation curves follow the same trajectory and remain close to each other, decreasing dramatically toward zero without significant deviation. The elastic net model performed worse than linear regression, as well, and k-nearest neighbors showed only a slight improvement. After tuning, the optimal value for the `n_neighbors` hyperparameter was determined to be 25, a relatively high number. This highlights a key limitation of the wine dataset: there is little meaningful connection between input features like acidity or alcohol content and the final quality rating.

These results show that wine quality cannot be reliably determined using only physicochemical attributes, at least with the models tested in this project. In contrast,

the Energy Efficiency dataset yielded much better results with the same models. Although the building features in that dataset were simulated rather than collected from real-world measurements, the data still suggests that machine learning is applicable for estimating energy efficiency based on building specifications such as relative compactness, surface area, wall area, roof area, and overall height. Meanwhile, attributes like orientation, glazing area, and glazing area distribution appeared to have a lesser impact on predicting heating and cooling loads.

Works Cited:

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